Rahul Chandra Nandagiri

Doctor of Business Administration Program, Midwest University, Wentzville, Missouri, United States

E-mail: chandrarahul416@gmail.com

The Editor-in-Chief, Knowledge-Based Systems

Dear Editor,

Please consider the enclosed Short Communication, “Aligning cybersecurity investment with business-impact risk: A knowledge-based AI decision-support model evaluated on real-world incident data,” for publication in Knowledge-Based Systems.

The paper presents KB-CIDS, a knowledge-based decision-support model that couples an explicit production-rule knowledge base (22 rules mapping incident patterns to nine control families aligned with CIS Controls v8 and NIST CSF 2.0), a calibrated machine-learning model of material business impact, and a Gordon–Loeb budget-allocation layer. Using 9,284 public incidents from the VERIS Community Database, all components were developed on incidents up to 2016 and evaluated out of time on 2017–2026. Business-impact-aligned allocation outperformed both undifferentiated and threat-frequency budgeting in all 18 sector–size profiles and closed 56% of the gap to an ex-post oracle, whereas threat-frequency allocation performed worse than an equal split. A component ablation is reported openly, including a negative result for imputing unknown impacts.

The work fits the journal’s aims of supporting human decision-making through knowledge-based and data-science techniques with applications in business, and its leading topics of intelligent decision support systems, knowledge representation and data-driven optimization.

The manuscript is original, has not been published previously and is not under consideration elsewhere. The data are public (CC BY-SA 4.0), and the complete code, derived data, figures and tables are openly available for reproduction. The use of generative AI tools in preparing the manuscript is declared in the dedicated section, and the author declares no competing interests.

Thank you for your consideration.

Sincerely,

Rahul Chandra Nandagiri